



OUTCOMES-BASED COURSE SYLLABUS

COURSE INFORMATION						
Course Code	CCOMP 010	Course Title	INFORMATION MANAGEMENT		Course Credit	3 units / 5hours
Course Description		The course introduces students the basic relational database concepts. During the course, students will learn how to design databases observing concepts and procedures in relational database design, document design using entity-relationship diagram, and use SQL to retrieve data and generate information. [Output] Students gain actual experience of database development from analysis of actual user requirements to implementation and testing of their database design. [Modality] The course is delivered through a combination of strategically designed face-to-face and online (synchronous and asynchronous) sessions.				
Pre-Requisites		COMP 006 -Data Structures		Co-Requisites		
VISION		A Leading Comprehensive Polytechnic University in Asia				
MISSION		▪ Advance an inclusive, equitable, and globally relevant polytechnic education towards national development. Strategic Goals Pillar 1: Teaching and Learning Pillar 2: Research and Extension Pillar 3: Internal Governance				
QUALITY POLICY STATEMENT		The Polytechnic University of the Philippines commits to provide inclusive and equitable quality education and promote lifelong learning opportunities for human advancement and social transformation through re-engineered polytechnic academic programs. Toward this end, we, the members of the PUP Community will vigorously and steadfastly endeavor to continuously improve the standard of university services to the satisfaction of our clients through the adoption and continuous review of our Quality Management System.				



STO. TOMAS CAMPUS GOALS	The PUP Sto. Tomas Branch commits to accomplish the following goals: 1. Sustain a culture of quality among the branch stakeholders by providing opportunities for student, faculty and staff development at the local, regional, national and international levels. 2. Develop and nurture collaborative research culture. 3. Champion knowledge and technology transfer through responsive sustainable and development-oriented community extension programs. 4. Provide a learning-conducive campus environment where technology driven resources are available to students, faculty member are available to students, faculty members and staff in enlarging their skills and competencies to meet the demands of the changing times. Ensure continuous improvement in the various areas of its operations through productive partnership and engagements with public, private and civil society acquire time
INSTITUTIONAL LEARNING OUTCOMES (ILO)	As a polytechnic state university, PUP shall develop its students to possess: 1. Critical and Creative Thinking. Graduates use their rational and reflective thinking as well as innovative abilities to life situations in order to push boundaries, realize possibilities, and deepen their interdisciplinary, multidisciplinary, and/or transdisciplinary understanding of the world. 2. Effective Communication. Graduates apply the four macro skills in communication (reading, writing, listening, and speaking), through conventional and digital means, and are able to use these skills in solving problems, making decisions, and articulating thoughts when engaging with people in various circumstances. 3. Strong Service Orientation. Graduates exemplify strong commitment to service excellence for the people, the clientele, industry and other sectors. 4. Adept and Responsible Use or Development of Technology. Graduates demonstrate optimized and responsible use of state-of-the-art technologies of their profession. They possess digital learning abilities, including technical, numerical, and/or technopreneurial skills. 5. Passion for Lifelong Learning. Graduates perform and function in society by taking responsibility in their quest for further improvement through lifelong learning. 6. Leadership and Organizational Skills. Graduates assume leadership roles and become leading professionals in their respective disciplines by equipping them with appropriate organizational skills. 7. Personal and Professional Ethics. Graduates manifest integrity and adherence to moral and ethical principles in their personal and professional circumstances. 8. Resilience and Agility. Graduates demonstrate flexibility and the growth mindset to adapt and thrive in the volatile, uncertain, complex and ambiguous (VUCA) environment. 9. National and Global Responsiveness. Graduates exhibit a deep sense of nationalism as it complements the need to live as part of the global community where diversity is respected. They promote and fulfill various advocacies for human and social development.





PROGRAM LEARNING OUTCOMES (PLO)		Alignment to ILOs								
		1	2	3	4	5	6	7	8	9
	1)Apply knowledge of computing fundamentals, knowledge of a computing specialization, and mathematics, science and domain knowledge appropriate for the computing specialization to the abstraction and conceptualization of computing models from defined problems and requirements.	✓			✓					
	2)Identify, analyze, formulate, research literature, and solve complex computing problems and requirements reaching substantiated conclusions using fundamental principles of mathematics, computing sciences, and relevant domain disciplines.	✓			✓					
	3)Apply knowledge and understanding of information security issues in relation to the design, development and use of information systems	✓			✓					
	4)Design and evaluate solutions for complex computing problems, its systems, components, or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations.	✓		✓	✓					✓
	5)Perform effectively as an individual and as a member or leader in diverse teams and in multidisciplinary settings.		✓	✓			✓		✓	
	6)Communicate effectively with the computing community and with society-at- large about complex computing activities by being able to comprehend and write effective reports, design documentation, make effective presentations, and give and understand clear instructions.		✓							
	7)Recognize the legal, social, ethical and professional issues involved in the utilization of computer technology and be guided by the adoption of appropriate, ethical and legal practices.			✓				✓		✓
8)Recognize the need, and have the ability, to engage in independent learning for continual development as a computing professional.				✓	✓			✓		
PERFORMANCE INDICATORS (PI)		Alignment to PLOs								
		1	2	3	4	5	6	7	8	
	• Simplify equations using various methods for logic simplifications. • Design, implement, test and debug a program, based on a given specification	✓								



	• Complete and successfully defend a thesis in line with the discipline. • Design and develop a computing solution using object-oriented approach • Implement solutions to circuit design problems using various logical gate functions. • Design a database based on user requirements using a widely used modeling notation		✓						
	• Design an overall architecture of the system and justify its appropriateness • Identify the strengths and weaknesses associated with different approaches to security. • Describe the set of controls and processes intended to protect information and information systems			✓					
	• Develop appropriate user interfaces for domain specific applications. • Evaluate how system and infrastructure components are organized into solutions in different environments.				✓				
	• Discuss with team members different software methodologies and techniques and identify best solutions for implementation. • Present a proposed solution in class or in a public forum • Justify the proposed solutions effectively through the use of scientific methodologies and document the results properly. • Analyze solutions employed by organizations to address different computing issues.					✓	✓		
	• Immerse/expose in an actual working environment in industry. • Propose an ethical and feasible software solution to the identified research problem that employs new designs, tools, and methodologies. • Evaluate professional, ethical and social issues of computing decisions.							✓	
	• Create a report on a conducted independent learning activity. • Identify the relevant/legitimate sources of technology based information and utilize them for continuous learning								✓
		Alignment to PLOs							



COURSE LEARNING OUTCOMES (CLO)									
		1	2	3	4	5	6	7	8
	1. Explain fundamental principles, concepts and evolution of database and information management systems			✓	✓		✓		
	2. Understand database models and structures.		✓						
	3. Create database design using logical, conceptual and physical database models						✓		✓
	4. Use SQL statements in creating and generating information required by the database users.	✓		✓	✓				
	5. Analyze solutions employed by organizations to address different database issues					✓		✓	
	6. Evaluate database design and formulate recommendations based on principles in database design. Document both assessments and recommendation							✓	

Legend: L – learn, P – practice, O – opportunity for development

OUTCOMES-BASED TEACHING AND LEARNING PLAN (OBTL PLAN)

Week (18 Week s)	Desired Learning Outcomes (DLOs)	Alignmen t to CLOs	Learning Content/Topics	Instructional Delivery Design			Assessment Tasks (TAs)
				Face-to-Face	Flexible Learning and Teaching Activities (FLTAs)		
					Synchronous	Asynchronous	
1	• Demonstrate a sense of readiness for the upcoming semester. • Identify their learning outcomes and expectations for the course.		Introduction to the Course a. Vision Mission Goals and Objective of the University, and College. b. Self-Introduction c. Course Overview	Discussion of the different learning content, grading system, and rules inside the f2f class	. Discussion of the different learning content, grading system, and rules inside the online class		

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	Recognize their capacity to create new understandings from reflecting on the course.		d. Grading System e. Classroom Management		Introductions Via MS Teams meeting		
2	Discuss the fundamentals and advantages of database, approaches to its design, and database structures/architecture.	CLO 1	Database Concepts - Fundamentals of Data, Information, Database, Data Warehouse and Data Processing Activities - Electronic Database (Range of Database Application, Components of Database Environment) - Database Approach (Advantages of DB Approach, DB Development Process, DB Models/Architecture)	Lecture and interactive discussions Video presentation	Lecture and discussions Video presentation Use of online tools / applications – mentimeter / mural,/ jamboard	Watch video of lecture Submit answers to questions provided Research on additional information on the topic discussed in the video.	Review Questions Home Work
3	Analyze forms, reports, and business rules and transform them into relations observing the properties of relational database.		Data Requirement Analysis - Analysis of Business Forms, Reports, Business Functions, - Analysis of Business Rules	Lecture and interactive discussions Recitation	Lecture and exercises Recitation via the chat box Assignment Exercises	Watch video of lecture Submit answers to questions provided Research on additional information on the topic discussed in the video	Graded Exercises
4-5	Design a conceptual database applying principles of relational		Conceptual Data Modeling using Entity-Relationship Model ERM	Interactive Discussions Case Modeling /	Lecture and exercises Recitation via the chat box	Watch video of lecture	Quiz 1 Graded Exercises on designing

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	database design and using standard notations and tools. • Use Entity relationship diagram as data model.		• ERD Symbols • How to create ERD	Diagramming Recitation	Assignment Exercises	Submit answers to questions provided Research on additional information on the topic discussed in the video	ERD Homework
6-7	• Design a relational database applying principles of relational database design and using standard notations and tools. • Use normalization to decompose relations with anomalies into well-structured relations. • Design relations in at least 3rd Normal Form		Relational Database Modeling (Properties of Relations, Integrity Constraints, Normalization)	Interactive Discussions Case Modeling / Diagramming Recitation	Lecture and exercises Recitation via the chat box Assignment Exercises	Watch video of lecture Submit answers to questions provided Research on additional information on the topic discussed in the video	Graded Exercises on normalization of database Assignment 2
8	• Utilize CASE tools to evaluate and implement database design. • Transform ERD to a logically equivalent set of relations in at least 3rd Normal Form		Logical Database Design (Transforming ERD into Relations)	Interactive Discussions Case Modeling / Diagramming	Lecture and exercises. Recitation via the chat box Assignment Exercises	Watch video of lecture Submit answers to questions provided. Research on additional information on the topic discussed in the video	Quiz 2
9	MIDTERM EXAMINATIONS						



10	- Understand Structured Query Language and statements. - Write DDL statements in creating database.		Creation of Database Using Data Definition Language	Interactive Discussions Actual Simulations & Demonstration	Lecture and exercises. Recitation via the chat box Assignment Exercises	Watch video of lecture Submit answers to questions provided Research on additional information on the topic discussed in the video	Hands-on Exercises on writing DDL Statements Hands-on Test
11-12	- Write DML SQL statements to generate information required by the database users using: - Simple SELECT Statement - SELECT Statement with WHERE clause (using Relational, Logical and Special database operators)	•	Data manipulation language (Basic SQL statements for data retrieval and simple manipulation)	Interactive Discussions Actual Simulations & Demonstration	Lecture and exercises Online Hands-on Exercised Assignment Exercises	Watch video of lecture Submit answers to questions provided Research on additional information on the topic discussed in the video	Hands-on Exercises on writing DML Statements Hands-on Test
13	- Write DML SQL statements to generate information required by the database users using: - SELECT Statement with Aggregate functions. - SELECT Statement with GROUP BY and HAVING Clause		Aggregate Functions	Interactive Discussions Actual Simulations & Demonstration	Lecture and exercises Recitation via the chat box Assignment Exercises	Watch video of lecture Submit answers to questions provided Research on additional information on the topic discussed in the video	Hands-on Exercises on writing Aggregate Functions Statements Hands-on Test



14-15	Write DML SQL statements to generate information required by the database users using multiple tables.		Complex Retrieval of Data (Use of multiple relations and subqueries)	Interactive Discussions Actual Simulations & Demonstration	Lecture and exercises Recitation via the chat box Assignment Exercises	Watch video of lecture Submit answers to questions provided Research on additional information on the topic discussed in the video	Hands-on Exercises on complex data retrieval using Join Hands-on Test
16-17	To apply the skills in database design and development		Development of Final Database Project	Group Hands-on / Database Development Period	Online Hands-on Activities	Creation of Paper and pencil database design	
18	PRESENTATION OF LEARNING OUTPUT (FINAL DATABASE PROJECT)						

**REFERENCES FROM THE NINOY AQUINO LEARNING AND LIBRARY RESOURCES CENTER (NALLRC)
OUTCOMES-BASED BOOK LISTINGS (CBBL)**

Elmasri, Ramez and Navathe, Shamkant B (2011). Fundamentals of Database Systems, Sixth Edition, ebook.

Bulusu, Lakshman (2008). Oracle PL/SQL: Expert Techniques for Developers and Database Administrators

Castillo, Melani L. (2017). Instructional Materials in Database Management System 1

Gavin Powell, Carol McCullough-Dieter (2007). Oracle 10g Database Administrator: Implementation and Administration

Thomas Connolly and Carolyn Begg, Database Systems, 3rd edition, 2007



Rajan, Claire (2007). Oracle 10g, Database Administrator II

OTHER REFERENCES

- www.ou.edu/class/aschwarz/Database/B02aChap2.ppt
- www.w3schools.com
- https://docs.oracle.com/cd/A57673_01/DOC/server/doc/SCN73/ch20.htm
- <http://www.vub.ac.be/BIBLIO/nieuwenhuysen/courses/chapters/client-server.pdf>
- <http://sqlfiddle.com/>
- <http://www.comentum.com/guide-to-Internet-database-development.html>
- <https://www.jdoodle.com/execute-sql-online/>
- <https://www.youtube.com/watch?v=zTs5zjSXnvs>
- www.codeblocks.com

Other Resources:

- Slide Presentation
- Sample Forms and Reports
- Computer System with DBMS-SQL Software/ Application

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CLASSROOM POLICIES (to be filled out by the assigned faculty)	
FACE-TO-FACE DELIVERY	FLEXIBLE TEACHING AND LEARNING ACTIVITIES (FLTAs)
General classroom guidelines - Students shall attend set contact schedule ready with all the materials and outputs required to be read, discussed, and/or submitted. Students should have also read required texts at least once before its scheduled discussion. - PLAGIARISM SHALL NOT BE TOLERATED. The following penalties will be strictly implemented to outputs proven to contain plagiarized words, phrases, clauses, sentences, paragraphs, or ideas: First offense – automatic failure in the output; Second offense – automatic failure in the output + letter from parent/s/guardian/s that acknowledges the offense; Third offense – automatic failure in the course. - Requirements shall be submitted on time. However, in special cases when students fail to submit requirements for some acceptable reasons, submissions will be subjected to deductions of no less than 0.25 per day. - Students who have any form of disability must inform the course instructor immediately so that alternative arrangements may be immediately considered. - All students are expected to read and strictly observe the PUP Student Code of Conduct https://drive.google.com/file/d/0B1BuDAuN0r8SX1BWx2N3N3FURzg/view?resourcekey=0-oi8IUy9PCFysh0FDyL5ipw Guidelines for the face-to-face delivery: - Strictly observe the minimum health protocols set by the university. - Check your schedule on the class Facebook page before going to school. - Be mindful of your classmates and teacher's time. Be alert, constructive, and responsive.	Synchronous sessions - Check your device ahead of your scheduled synchronous meeting (camera, microphone, keyboard, speakers, etc.) - Attend the synchronous class on time. - Be ready to turn on your microphone and camera anytime. - Choose a comfortable space to attend the online class. - Click the 'raise hand' button and wait to be acknowledged by the teacher(s) before unmuting your microphone. - Do not abuse the chatbox. - Read the assigned materials before attending the class. - Be mindful of your classmates and teacher's time. Be alert, constructive, and responsive. Asynchronous sessions - Study the sections and functions of the assigned learning management system (LMS) ahead of time. - Check the expected submission/turn in schedule at all times. For some timed activities, late submission may cause deductions to your grades. For group activities, discuss the best time and platform to discuss the assignment of tasks with your groupmates. - Ask for help from your teacher(s) and classmates when necessary. (Follow the rules on sending an effective email to your teacher. A separate discussion shall be allotted for this.)



COURSE REQUIREMENT/S with CLO links	GRADING SYSTEM																				
• Participation in the classroom activities (CLOs 1 and 2) • Assignment and Asynchronous Exercises (CLOs 1 and 3) • Laboratory Hands-on Exercises (CLOs 3-6) • Midterm Examination - (CLOs 1 and 4) • Oral Presentation - 35% of the Final Examination, to be graded using rubric for Database System Presentation (CLOs 3-6)	Class Standing Attendance & Class Participation Assignment Asynchronous Exercises/ Activity Hands-on Exercises 70%																				
	Midterm / Final Examinations 30% 100%																				
	Midterm Grade + Final Term Grade = Final Grade																				
Rubrics for Assessment (to be filled out by the assigned faculty)																					
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CLASS INFORMATION		FACULTY INFORMATION																			
Section: BSIT 2-1 / BSIT 2-2		Name of Faculty: DR. MELANI L. CASTILLO																			



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Time: MTH 9:00-12:00 (BSIT 2-1) / T/F 1:00 -4:00 BSIT 2-2	Consultation Time: Mon & Thurs 5:00 - 6:00
Room: COMLAB 2	Office Tel. No./ Mobile Phone No. 09258232420
Semester: Second Semester, AY 2024-2025	Institutional Email: mlcastillo@pup.edu.ph

Prepared by:	Reviewed and checked by:	Recommending Approval:	Approved by:
 <u>Dr. Melani L. Castillo</u> Faculty Date: <u>2/6/26</u>	<u>Engr. Liza Marie B. Nuevo,MSEE</u> Head, Academic Programs Date: _____	<u>Dr. Armando A. Torres</u> Director Date: _____	<u>Prof. Pascualito B. Gatan</u> Vice President for Campuses Date: _____

Form No. _____
Revision No. _____
Date of Approval _____
Semester _____
Academic Year _____

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